Mind Guard AI

Problem Understanding

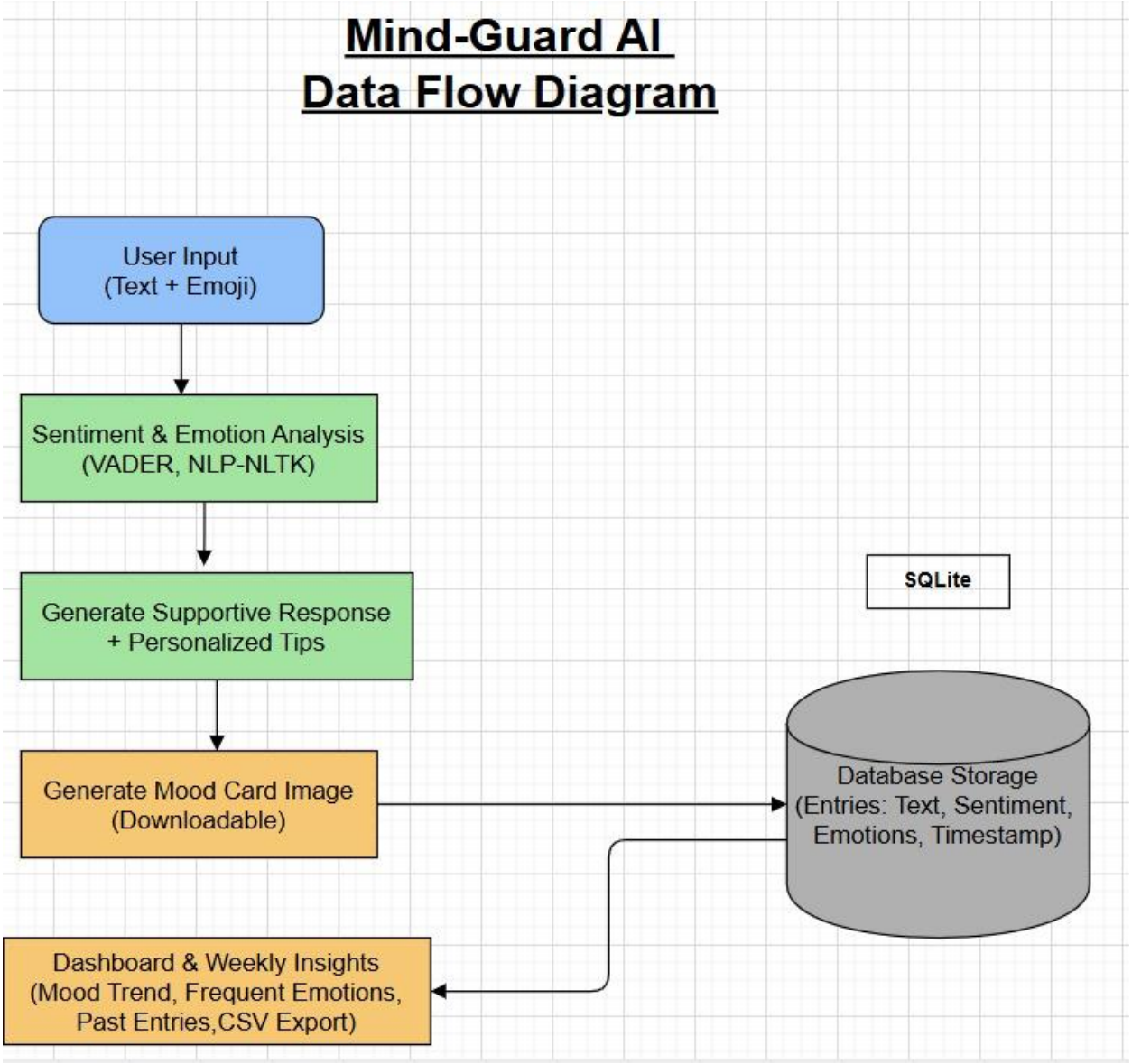
Mental health is an essential aspect of overall well-being, but people often struggle to track their moods and emotions daily. Traditional journaling can be tedious, and understanding one's emotional patterns is difficult without analysis.
Mind Guard AI addresses this problem by allowing users to input their daily thoughts and moods, automatically analyzing them for sentiment and emotions, and providing personalized supportive messages and coping tips.

Approach

1. **Input:** Users can submit text entries or voice recordings describing their thoughts. Voice inputs are converted to text using speech-to-text processing. Users can also select an emoji representing their mood.
2. **Text Analysis:** Natural Language Processing (NLP) techniques are used to:
    - Tokenize and lemmatize text
    - Detect emotions (Joy, Sadness, Anger, Fear, Anxiety, Surprise)
    - Analyze sentiment using VADER sentiment analyzer
3. **Emoji Integration:** The selected mood emoji is converted into a numeric score and combined with text sentiment to improve accuracy.
4. **Output:**
    - Generate supportive messages tailored to the user's mood
    - Suggest coping tips based on detected emotions
    - Create shareable mood cards (image visualization of user's entry and analysis)
5. **Dashboard & Trends:** Track daily streaks, visualize mood trends, analyze weekly insights, and export history as CSV.

Challenges Faced

- Handling negations in text (e.g., "not happy") for accurate emotion detection.
- Combining text sentiment with emoji mood ratings to create a unified score.
- Dynamically generating mood cards with variable text length and ensuring text wraps correctly.
- Calculating streaks and trends from database entries while handling missing days.
- Ensuring real-time visualization of mood and emotions on the dashboard.

What I Implemented:

- Full backend logic in Python for sentiment and emotion analysis.
- Integration of voice input using speech-to-text functionality.
- Integration of emoji mood ratings with NLP analysis.
- Dynamic mood card image generation using Pillow.
- Dashboard with trend plots and weekly insights using Pandas and Matplotlib.
- CSV export functionality for user history.
- Designed the Flask web interface for text input, mood selection, and viewing results.